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The Use of Chatbots in Energy Education

Othman BENMALEK[1,a *], Nasr-eddine JABARI[1,b], Khaoula BENMOUSSA[2,c], Moulay Driss EL OUADGHIRI[3,d]

[1]Faculty of Sciences, Moulay Ismail University, Meknès, Morocco

[2]Research Laboratory in Educational Sciences, Philosophy and Humanities, Higher Normal School, Moulay Ismail University, Meknes, Morocco

[3]IA Laboratory, Faculty of Sciences, Moulay Ismail University, Meknes, Morocco

[a]o.benmalek@edu.umi.ac.ma, [b]jabarinasredinne@gmail.com, [c]K.benmoussa@umi.ac.ma, [d]dmelouad@gmail.com

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Abstract. Since the use of artificial intelligence (AI) in educational contexts has increased rapidly, there has been a surge in the use of chatbots as interactive agents of learning in education. Recently, chatbots have shown great promise for providing engagement, access, and personalized learning in a variety of academic and non-academic fields. In the area of energy education, there are unique benefits that can be found because energy education is an applied, interdisciplinary, and technical area of scholarship with a focus on sustainability and climate change. This literature review examines the current literature regarding the use of chatbots in education and explores the potential uses of chatbots for energy education. Some of the benefits provided by chatbots include increased access and personalized learning methods, while challenges related to chatbots include limited levels of reasoning ability, reliability of data, and ethical considerations. Furthermore, a technical case study demonstrates the creation of a Retrieval Augmented Generation (RAG) chatbot using 267 curated documents for Energy Education. Initial testing (n=6) achieved 100% accuracy with instantaneous response times, showcasing the potential for providing domain-specific answers on topics related to the renewable energy sector, efficient use of energy, and sustainability. The literature review concludes with a focus on future improvements for using chatbots to provide enhanced levels of natural language processing (NLP) and adapt to learner needs in order to enhance energy literacy and sustainability education.

Introduction

The continually evolving nature of education resulting from advances in technology has created an unprecedented opportunity for us to reconsider the role that the educator plays in a learner's educational experience. We can now understand how to create new boundaries for our current practices regarding these teaching practices. While traditional classroom-based habitual approaches to teaching have overall been beneficial and have produced successful results for most students; these types of instructional strategies are seldom able to consistently optimize all aspects of the learning process and the educator's ability to deliver tailored educational support on behalf of each learner is largely absent, therefore leading learners to experience an incomplete, non-optimized learning experience. Ironically, the global energy transition and climate crisis is bringing attention to the extreme importance of energy education, giving some base for students and citizens to comprehend the attributes of gaining an education and the various ways of energy production, sustainable energy use, and energy conservation.

Chatbots[1] are automated conversational agents, which are increasingly being utilized as a key way to fill gaps in student learning[2]. Chatbots[3] can normally engage in conversation by

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replying instantly, supporting students through complex challenges, and providing interactive rationales that allow learners to develop knowledge in an explanatory and effective manner. In regards to energy education specifically, chatbots could initiate language to break down technical concepts effectively, provide immediate answers to student questions, facilitate the expression of ideas or beliefs on the part of students at any time, and create a framework to allow students to explore engagement in an active learning capacity.

This paper is structured as a review study examining existing literature on chatbots in education, with a specific focus on energy education. In addition to the literature analysis, a technical case study based on a retrieval-augmented generation (RAG) chatbot is presented to illustrate practical implementation.

Review and Screening Methodology

Literature Review. Recently, Ai chatbots have been shown to be effective tools in education. They support student engagement in learning, provide adaptive feedback and personalized learning within formal educational systems, and create opportunities for interaction that increase student motivation while reducing resistance to educational technologies. These findings highlight the effectiveness of conversational ai systems as supportive educational tools.[4]

Most studies to date, focus on general education rather than a more domain-specific education. Energy education, involves interesting systems such as renewable technologies, grid dynamics and sustainability concepts; still lacks empirical work on AI integration. AI chat systems[5] are making energy concepts more accessible to learners, from novices to professionals. These systems addresses a new way of learning and retaining information. [6]

Additionally, there are multiple studies exploring generative AI assistants in energy and sustainability tasks that demonstrates that chatbots can assist with real-world reasoning. These tools act interactively by giving responses that aligns with the goals of energy education where complexity and interdisciplinarity are central challenges.[7]

This emerging evidence states that knowledge based chatbots, especially those using retrieval-augmented generation (RAG) established based on energy datasets, can bridge gaps between traditional content and interactive learning; In the following sections, a retrieval-augmented generation (RAG)-based chatbot for energy education is described and evaluated.

Chatbots in Education. Education-based chatbots[8] are designed for three primary purposes: information, practice, and engagement. They provide information to students by responding to their questions and helping students clarify difficult concepts. They provide practice by giving quizzes and exercises with immediate feedback. They provide engagement by helping to simulate conversations that lead to a more casual and approachable disposition toward learning.

There are important benefits to leveraging chatbots in the educational context[9], such as:

Availability: A learner can access support any time - this leads to reduced reliance on teacher availability.

Personalization: Chatbot responses can be personalized based on learner progress or a student's previous interactions.

Scalability: One chatbot could be used by a large cohort of students at the same time.

Nonetheless, chatbots have many disadvantages to consider. Some examples of these limitations are the inability of chatbots to be able to think through complex topics like a human instructor can, chatbots sometimes respond with partial information, and chatbots are completely reliant on the standard of their training materials. These weaknesses must be evaluated and corrected in order for chatbots to reach their fullest potential; particularly in the area of specific academic topics, such as education in renewable[10] sources of energy.

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The Role of Chatbots in Energy Education. Energy education[11] involves a range of topic areas, including renewable energy sources, energy efficiency, climate change, and sustainability. Many of these topic areas are complex, combining scientific and technical areas with socio-economics. Students can struggle with scientific or technical language, conceptual or abstract models, and applying conceptual learning to the real world.

Chatbots could assist in making energy education more accessible and interactive. They could:

Explain things: for example, provide a simple layperson's analogy to the process of converting the energy of the sun into electricity in solar panels.

Facilitate step-by-step guidance: for example, provide a learner with step-by-step instructions to solve problems or to explain energy-related case studies.

Facilitate awareness campaigns: encourage sustainable behavior by responding to questions, about energy, energy in homes, energy networks, etc., that people encounter every day.

Facilitate simulations: act as a reminder for students, explain protocols, or clarify things quickly.

In providing this type of support people not only learn better, but they also are likely to be more actively engaged.

Findings and Discussion

Proposed RAG Chatbot System. Dataset & preprocessing. For our study, we have used a domain-specific energy knowledge corpus from the *EnergyEducation encyclopedia.* This website contained a lot of chapters that has a lot of definitions about energy related terms. We scraped the website using a python Library called *BeautifulSoup*, extracting all relevant pages and chapters which reached a total of *300 pages*. Each entry was stored in a *Json file* containing three fields: *URL, Title and Text*. While inspecting the collected Dataset, we have found that some images were taken in count as terms which were deleted as part of preprocessing because only clean text is relevant for text-based retrieval. This preprocessing step resulted in high quality of *267 pages*, structured dataset ready for downstream embedding and indexing.

Embedding & Indexing. The structured dataset is represented by text chunks; In this step, each text chunk was embedded using the *MiniLM all-MiniLM-V6-v2* sentence transformer, which produces 384-dimensional vector representations of text. The embeddings capture semantic captures similarity between queries and knowledge chunks. These vectors were indexed using *FAISS (Facebook AI similarity Search),* a library optimized for fast similarity search over large dense vectors. The Faiss index allows retrieval of top-k most relevant chunks given a user query which facilitates efficient, fast and accurate access to knowledge corpus. Both the embeddings and the index were stored locally, enabling reproducible and a system that can be optimized very fast without reliance on external APIs.

RAG pipeline. The RAG[12] system operates in three stages. First, a user question is encoded using the sentence-Transformer used for the dataset. Then, the FAISS index retrieves the top-k most semantically relevant text chunks. Finally, these chunks provide contextual information for a language model to generate answers. A schematic overview of the RAG pipeline is presented in Fig. 1, illustrating the flow from user input to final generated answer**.**

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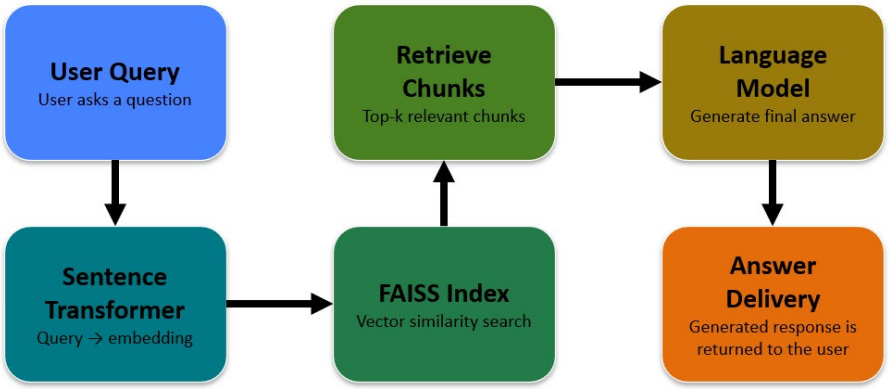


Fig 1: RAG architecture for the proposed energy education chatbot

Test Examples. To assess the system's performance, we tested a set of representative queries related to energy concepts, as presented in Table 1.

Table 1: Chatbot's responses based on the user questions

Question	Chatbot Response	Retrieved Topic Domain
What is solar energy?	Solar thermal power plants are electricity generation plants that utilize energy from the Sun to heat a fluid to a high temperature. This fluid then transfers its heat to water.	Solar Energy
Explain wind power generation	Wind power is the generation of electricity from wind. Wind power harvests the primary energy flow of the atmosphere generated from the uneven heating of the Earth's surface by the Sun.	Wind Energy
What are fossil fuels?	Fossil fuels are fuels made by geological processes acting on ancient dead organisms buried hundreds of millions of years ago.	Conventional Energy
What is energy efficiency?	Efficiency in physics is a comparison of the energy output to the energy input, representing how effectively energy is converted into useful work.	Energy Efficiency
What are electric vehicles?	Electric vehicles are vehicles that use electric motors for propulsion and onboard electricity storage systems, producing zero tailpipe emissions.	Sustainable Transport
What is mass?	Mass is a measure of how much matter is contained within an object and is expressed in kilograms in the SI system.	Physics

Practical Applications of Chatbots in Energy Education. We can look at the advantages chatbots may afford the field of energy education by examining their potential uses in the real world. Some examples of how a chatbot could be utilized to provide relevant information and assist students include:

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Teaching Assistant: Chatbots may serve as a teaching assistant when responding to frequently asked questions from students, explaining concepts from lectures, or providing supplemental material that would not place an undue burden on educators. For instance, if students had questions about solar, wind, or hydro technologies in the renewable energy module, asking a chatbot for definitions, examples, or summaries would not negatively impact their understanding of the content.

Self-Directed Learning: For course participants enrolled in asynchronous (i.e., internet) or distance education/training, chatbots could accompany them through their course modules, recommend supplemental materials and provide immediate completion of quizzes, followed by grading and feedback. Ongoing education will be better facilitated outside of traditional classroom settings.

Promotional Use: Chatbots can be developed for the general public to raise public awareness and share information related to renewable/sustainable energy. For example, a chatbot could be created to address FAQs related to decreasing home energy use, selecting energy-efficient household appliance options, and understanding the national renewable/sustainability policy issues.

Academic Preparedness: With regards to vocational educational programs and industrised training, all bots can be adjusted to develop learners' skill sets and provide employees with industry safety protocol, maintenance of energy apparatus, and education on/ upskilling to adapt to new energy technologies. Bots can therefore serve as a means of conducting education and providing a professional environment to enable learners take advantage of various learning and/or educational opportunities in the workplace.

Such implementations illustrate that bots are applicable to theoretical ends, but can also serve as practical tools that improve learners' experience, stimulate interest in learning, and enable learners to attain mastery over energy-related issues.

Discussion. Chatbots as educational tools present potential as well as challenges in the application of chatbots to educational applications relevant to the field of energy. Access to energy information on-demand regardless of the time of day or where you may be located is one of the main benefits of deploying chatbots as educational aids. By providing on-demand access to energy knowledge, students are able to take a self-directed approach to their learning; they can control how much time they spend on a concept and how deep into a topic they wish to go, depending on their specific needs as learners. This function of a chatbot will come in particularly useful when educating students about energy topics, as many students desire repeated exposure to more difficult concepts related to renewable energy systems, energy efficiency, and sustainability.

The results that were generated from the assessment of the proposed RAG chatbots have demonstrated how effective the use of RAG technology is at providing domain-specific information about energy concepts. The examples of the queries and subsequent responses that were provided in the study show that by using a RAG model, the chatbot can develop informed and appropriate answers to energy-related questions by pulling information from an existing database of energy resources. This aspect of the RAG system provides reliability to students by ensuring their responses are based on concrete references and eliminates or significantly reduces the likelihood of providing inaccurate or incomplete answers by chatbots that solely generate responses without referencing an external data source.

While the RAG architecture increases the accuracy of generated responses, it does not assure that all generated answers are accurate. The quality of the answers and their validity is still dependent on whether the data used to create those responses contain current, relevant, and complete information. There is the possibility that due to rapid advances in Energy Technology,

the data within a chatbot may become outdated unless it is regularly updated with new information from a knowledge base

As demonstrated by these evaluations, the chatbot provides generic or less relevant answers to questions that are out-of-domain. This is to be expected because the system is designed to return only the type of answer it is trained to give based on its training set. It is a limitation of the system, but also serves as a safeguard. It ensures that the chatbot's responses remain within the pedagogical boundaries of the original intent of the system, i.e., education.

This brings up an important question about how this type of technology will evolve and whether or not it might take over some of the functions of a teacher. The goal of this type of system is not to replace a teacher, but to augment the traditional teaching process by acting as an additional learning resource. Teachers and instructors remain the primary resource for guiding learners and validating the accuracy of information presented to learners. One of the more important aspects of designing a chatbot is ensuring that it clearly communicates the nature of its limitations and the sources from which it retrieves educational materials to learners.

In the future, Improved Natural Language Processing (NLP) technology combined with improved Machine Learning (ML) technology will likely increase both the ability of chatbots to better understand content and respond to learner questions in a more personalized context, while maintaining content validity through retrieval-based grounding.

As a whole, RAG-type chatbots are a viable way to provide an educational platform for the subject of Energy and meet considerations around ethics, data set maintenance and pedagogical integration[13]. The key considerations of this research are highlighted through their findings.

Conclusion and Perspectives

Conclusion. Chatbots are at the frontiers of energy education. As an innovative combination of the key elements of Access/Personalization/Engagement, they provide students with essential and relevant educational resources in a complex field like Renewable Energy, Efficient Energy Use and Sustainability.

Although there are still some challenges regarding the accuracy and level of detail in Chatbot responses, their use as a collaborative classroom tool or educational resource has already proven to offer tremendous value for creating an effective educational environment. Chatbots can help encourage learners, identify and correct areas of deficiency in traditional forms of instruction, and align broad-based educational goals with the principles of Sustainability and Energy Literacy.

In essence, Chatbots should be viewed as a tool that can supplement and build upon the overall learning ecosystem, rather than diminishing the impact of the teacher in the learning process.

Perspectives and Future work. In the future, we will continue to increase the size and diversity of the Energy Encyclopedia's underlying Knowledge Base by establishing links to other authoritative energy encyclopedias, open educational resources, and institutional repositories. A larger volume and diversified set of curated data will increase retrieval quality and response relevancy for advanced energy topics. A third avenue of future development is the provision of a dedicated, purpose-built user interface that will facilitate the usability and interaction of learners with the educational materials contained in the system through features such as guided query functions, topic navigation, and contextual feedback.

Future improvements might also include regular updates to the Knowledge Base with new content, integration with Learning Management Systems, and user-centered evaluations of the system's pedagogical effectiveness in actual learning environments.

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